Xavier Bruant

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EDUCATION

Mechanics PhD, LMS/PMC – École Polytechnique *(Palaiseau, France)* **2023 – 2026**[1]
Chemo-mechanical couplings in Si-based electrodes for Li-ion batteries,

- Modeling of the rheological behavior of lithiated silicon, experimental validation via stress and voltage measurements.
- Modeling of the phase transformation occurring during the first lithiation of a silicon thin-film electrode, FEM implementation with an Arbitrary Lagrangian-Eulerian approach and experimental validation.
- Hybrid phase-field modeling for finite-strain problems with nonlinear kinetics, FEM implementation.

Advised by: Laurent Guin, Michel Jabbour and Mathis Plapp.

Space Engineering MS, Caltech *(Pasadena, CA, USA)* **2022 – 2023**
Solid and Fluid Mechanics, Space Engineering, Applied Mathematics.

Ingénieur Polytechnicien, École Polytechnique *(Palaiseau, France)* **2019 – 2023**
Mechanics, Physics, Applied Mathematics.

Classe Préparatoire PT, Lycée Sainte-Geneviève *(Versailles, France)* **2017 – 2019**
Two-year university-level intensive program in engineering sciences, physics and math to pass competitive nationwide exams.

RESEARCH EXPERIENCE

Visiting PhD student, Brown University *(Providence, RI, USA)* **Spring 2025**
Experimental work in Pr. Pradeep R. Guduru's group on the chemo-mechanical behavior of silicon thin-film electrodes,

- Silicon thin-film electrodes manufacturing,
- Stress measurements via curvature monitoring with the Multi-Beam Optical Sensor technique.

Research Intern, M&M Team – ETHZ *(Zurich, Switzerland)* **Summer 2022**
Theoretical and numerical modeling of ferroelectric materials in the Mechanics and Materials team led by Dennis Kochmann,

[1] (expected)

- Numerical simulation of a stochastic Allen-Cahn phase-field model of ferroelectric switching in tetragonal ceramics with an FFT-based solver,
- Analytical computation of Hadamard jump conditions to validate the diffuse-interface numerical model,
- Data post-processing and segmentation algorithms for domain-wall detection.

R&D Intern, STMicroelectronics *(Rousset, France)* **Summer 2021**
Numerical simulations for phase-change memory design optimization,

- Numerical simulation of a drift-diffusion model for coupled conduction and heat transfer in phase-change memory devices.

TEACHING EXPERIENCE

Teaching Assistant, École Polytechnique *(Palaiseau, France)* **2023 – 2026**
Teaching and grading in the following classes: Introductory Physics (bachelor, 1st year - teaching), Solid Mechanics (bachelor, 3rd year - teaching), Computational Fluid Dynamics (Graduate level - grading), Computational Solid Mechanics (Graduate level - teaching), Fluid Mechanics (Graduate level – tutoring).

Oral Examiner, Lycée Sainte-Geneviève *(Versailles, France)* **2020 – 2021**
Spent two hours a week to examine three sophomores during an hour-long oral examination on engineering (mechanics and control systems).

"AWARDS"

- **Best Presentation Award**, GFECI (Société Chimique de France) **2024**
- **Monge scholarship**, École Polytechnique **2023 – 2026**
- **Research Internship Award**, École Polytechnique **2022**

RELEVANT RESEARCH SKILLS

Modeling:

- Non-equilibrium thermodynamics and multiphysics couplings (Transport, Phase transformations, Electromagnetism, Electrochemistry),
- Solid and Fluid mechanics,
- Free-boundary problems (Phase Field and Arbitrary Lagrangian-Eulerian approaches)

Numerical Simulation:

- Computational mechanics methods (Finite Elements, Finite Volume, Finite Difference methods),

- Applied Mathematics (Functional analysis, Convex optimization, Bayesian Statistics),

IT: Python (FenicsX, JAX), LaTeX, Matlab, R.

Languages: French (mother tongue), English (advanced, 111 TOEFL), German (basic knowledge).

PUBLICATIONS

- Indergand, Bruant, Kochmann, *Domain pattern formation in tetragonal ferroelectric ceramics*, JMPS, 2023.
- Le, Bruant, Phung, Ozanam, Rosso, Guin, *Stress dependence of the chemical potential of lithium in a silicon electrode*, JMPS, 2025.
- Bruant, Le, Ozanam, Rosso, Guin, Jabbour, *A sharp-interface model for the two-phase lithiation of silicon electrodes*, in preparation.

CONFERENCES AND TALKS

- *Effect of Stress and Viscoplastic Behavior on the Electrochemical Response of Amorphous Silicon Electrodes*, EMMC, 2024.
- *Chemo-mechanics of Phase Transformation in Amorphous Silicon*, ESMC, 2025.
- *Chemo-mechanics of Two-phase Lithiation in Amorphous Silicon*, EMMC, 2026.
- *A phase field model with nonlinear kinetics for the irreversible lithiation of silicon electrodes*, PFMCS (Workshop on Phase Field Modeling of Complex Systems), 2026.
- *Modeling and Simulation of the Two-phase Lithiation of Amorphous Silicon*, ECCOMAS, 2026.